

# Some advantages and disadvantages of coding types

| Coding types        | Advantages                                                                                        | Disadvantages                                                                        |
|---------------------|---------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| Imperative          | Easy to reuse code sections to simplify code                                                      | Higher risk of errors than declarative                                               |
| Declarative         | Clearer than procedural / imperative                                                              | Might be difficult to understand for someone who is not a domain-specific specialist |
| OOP                 | Good for making sophisticated software                                                            | Steep learning curve                                                                 |
| Functional          | Functions are easier to understand because they do not affect any variables outside of themselves | The type of function used can reduce the readability of the code                     |
| Procedural          | Easy to track the flow of a program                                                               | Can be hard to write and far-abstracted from reality                                 |
| Logic programming   | High level                                                                                        | Difficult to apply in the context of more general computation                        |
| Parallel processing | Better efficiency so can achieve things faster than with sequential programming                   | Can be complex to write                                                              |
| Database            | Direct clear statements that are structured similar to human sentence structure                   | Does not have the functionality of imperative languages                              |